

CoRoT-2b

R-0078

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_240419 · created 2026-07-15T22:12:00+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460419.9428 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.223 +/- 0.065
Transit depth (Rp/Rs)^2	4.97 +/- 2.89 [%]
Orbital Inclination (inc)	88.5 +/- 2.07
Airmass coefficient 1 (a1)	1.01 +/- 0.82
Airmass coefficient 2 (a2)	-0.1 +/- 0.32
Scatter in the residuals of the lightcurve fit is	86.2 %
Best Comparison Star	#1 - [274, 244]
Optimal Aperture	1.56
Optimal Annulus	4.74
Transit Duration (day)	0.0976 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.223 $\pm$ 0.065 vs 0.1667 (deviation 0.76 σ)
Duration fit vs published	0.0976 d vs 0.095 d (deviation 0.37 σ)
Mid-time O-C	-5.8 min
Depth SNR	3.0
Residual scatter	86.2 %

Light curve

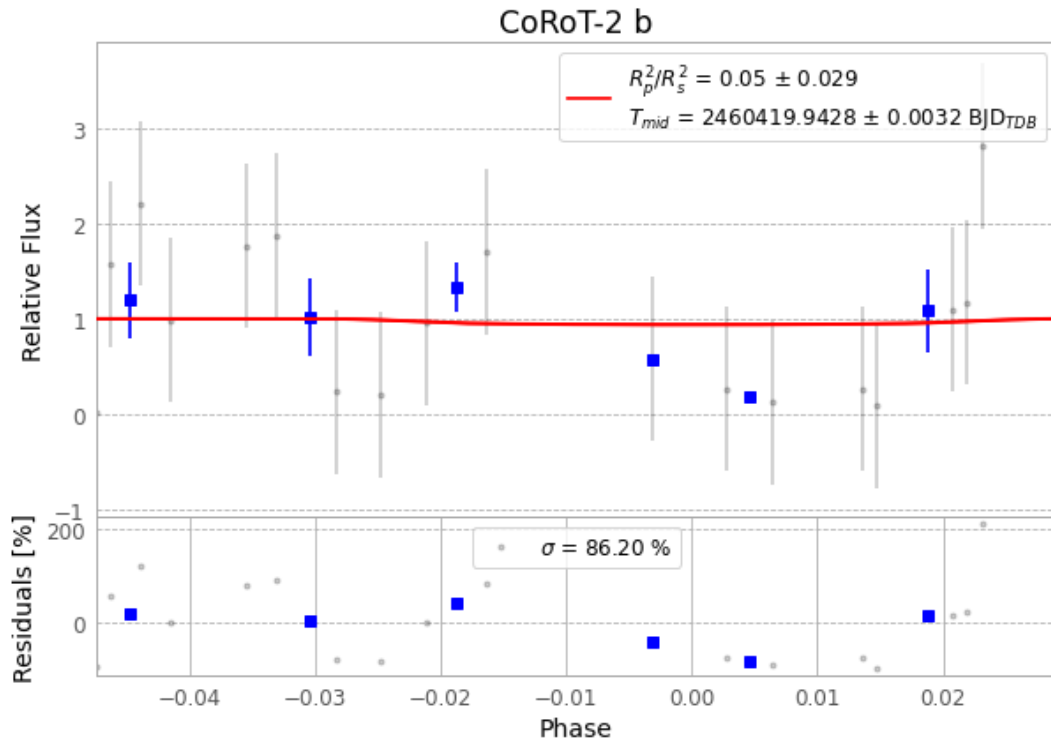


Figure 1 — FinalLightCurve_CoRoT-2 b_2024-04-19.png (SHA-256 369f3543c15ca6d0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [404, 232], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 10880af47bda1d52...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

67 FITS frames (44 MB), CoRoT-2240419083615.FITS → CoRoT-2240419115415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-240419.log (6f7a67c46a9d...), FinalLightCurve_CoRoT-2 b_2024-04-19.png (369f3543c15c...), FinalLightCurve_CoRoT-2 b_2024-04-19.csv (a8e24be35f1e...)

Compute resources

Wall time 295.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 22:12Z.